

Saffron Stigma Separator

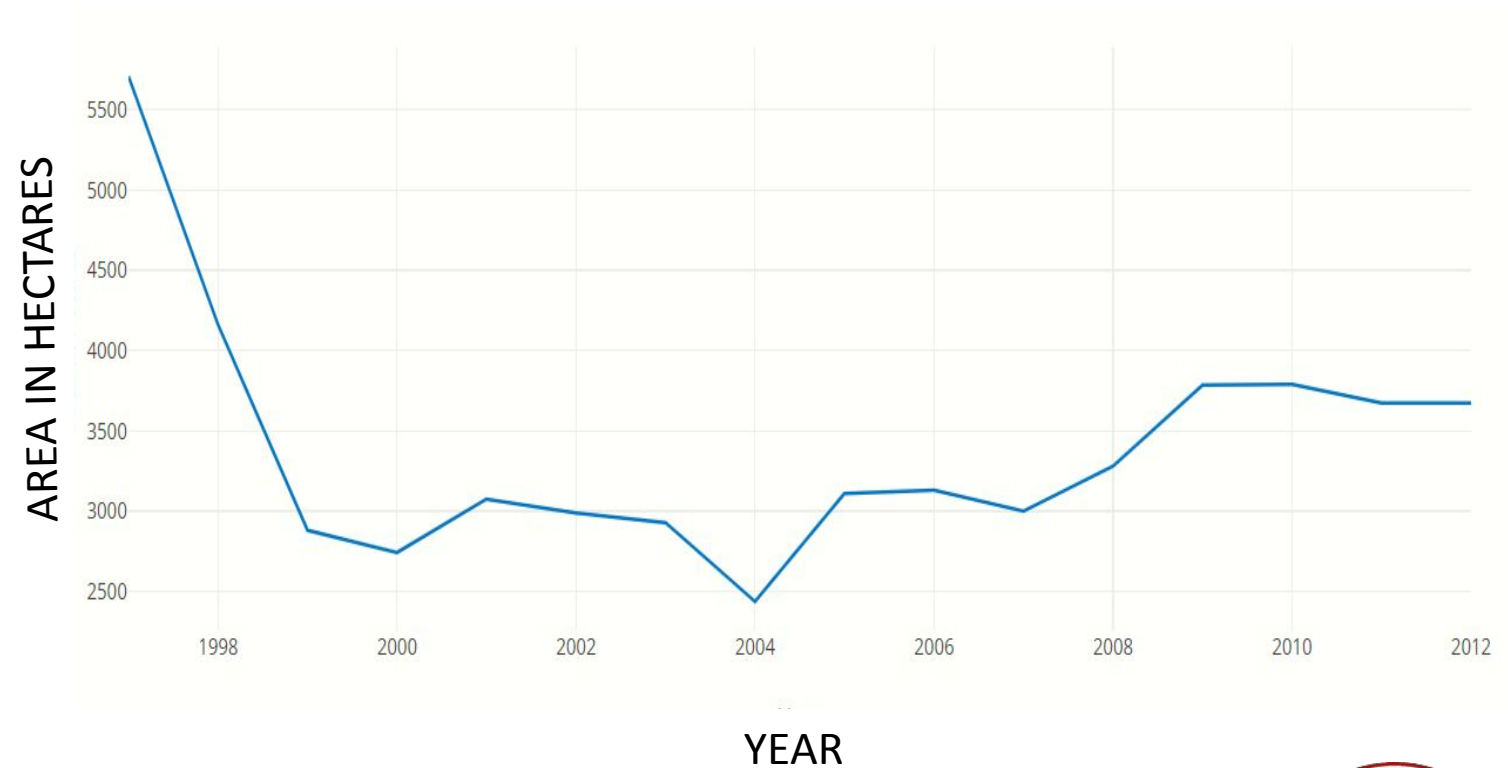
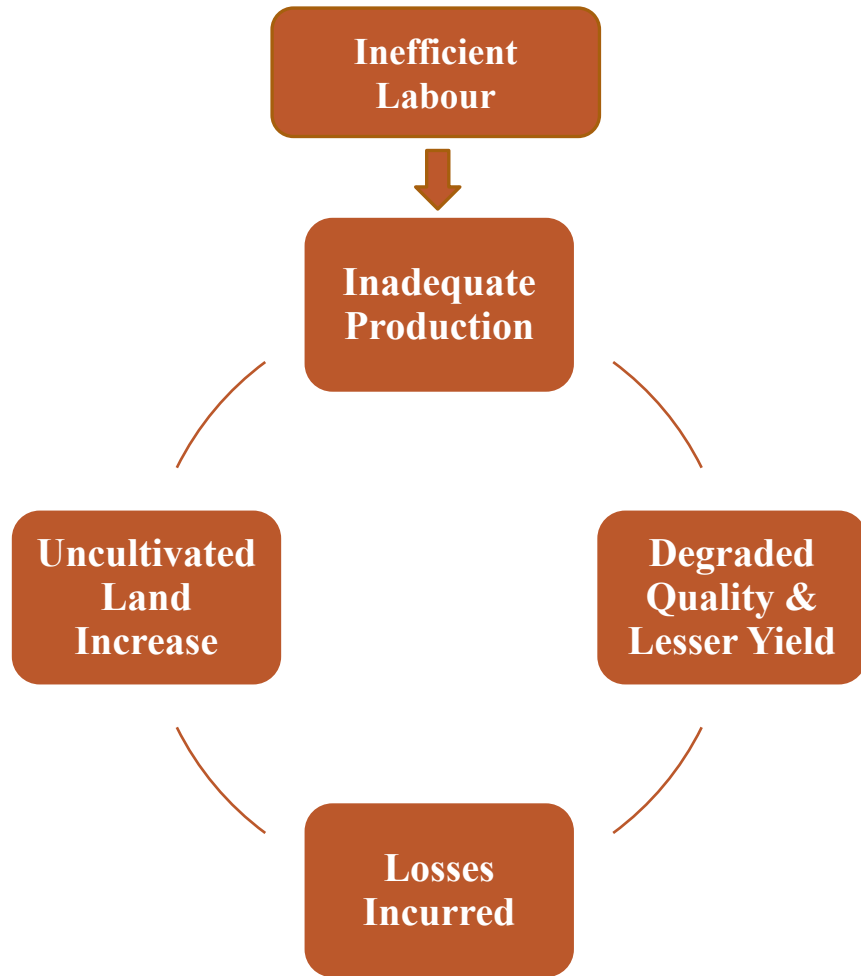


Why Saffron?

- Ø Saffron is lighter than a feather but packs a hefty wallop when we come to sell !
- Ø Highly Labour Intensive Crop: affects it's Cost
- Ø Lack of Mechanization
- Ø **Intelligent Automation** can be the 'Break Point'

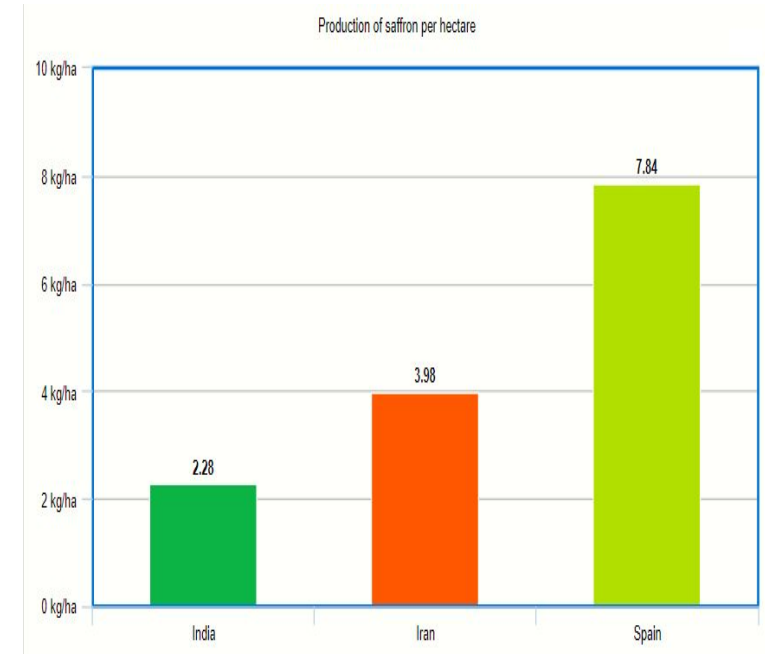


Present-Day Scenario



Present-Day Scenario

- Ø India's Saffron is of the finest quality in World and is of great importance in International Trade
- Ø Recognized by the Central Government: Initiated "**National Saffron Mission**".
- Ø Mechanized solutions not implemented in India.
- Ø Scientific approaches can lead to boost in productivity as in Spain.



The Problem is 'DRUDGERY' !

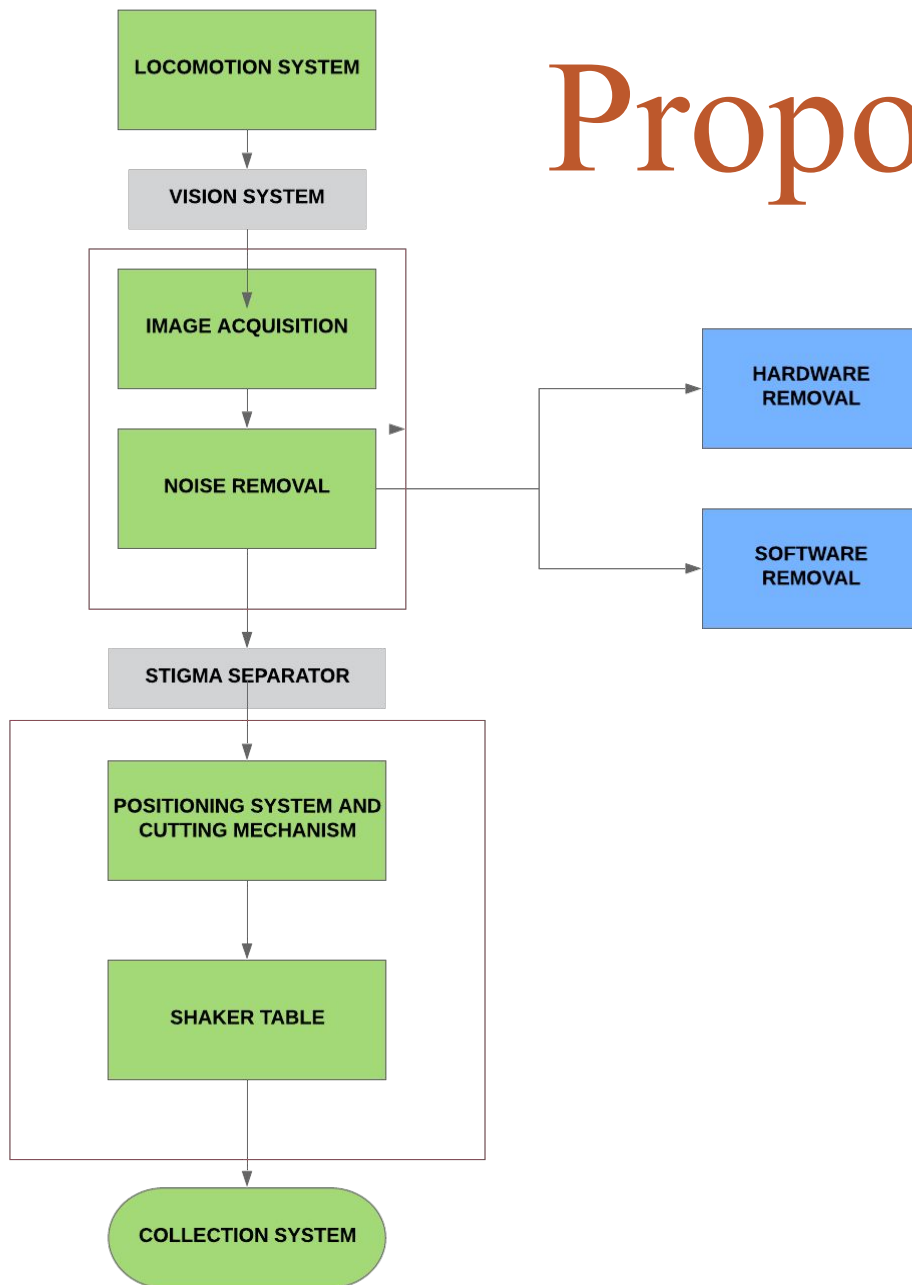
- Ø Stigma Separation involves intensive labour
- Ø Labour contributes significantly to the cost
- Ø Harvest Season: **200 hours** of rigorous labour
- Ø A hectare of field demands 10-15 labourers
- Ø Timely Separation: 400% increased output*

Duration of Separation	Recovery Rate
10-12 hours	37g/Kg of Flowers
36-72 hours	7g/Kg of Flowers

*According to the manual on "Good Practices for Saffron Production in India" issued by Sher-e-Kashmir University of Agricultural Sciences and Technology of Kashmir.



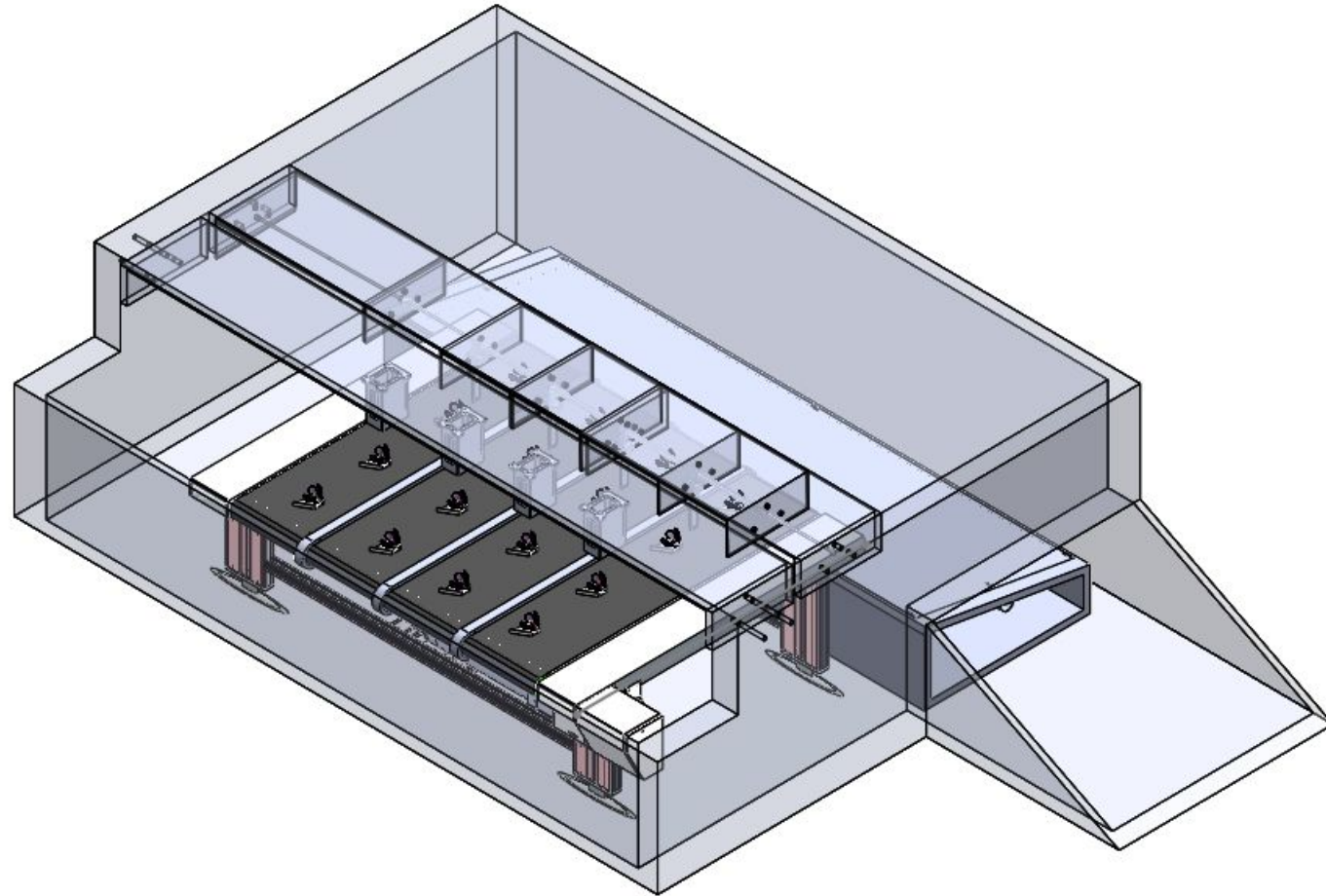
Proposed Solution



- Ø Stigma Separation: Repetitive in Nature
- Ø Involvement of Labour is Redundant
- Ø Process requires Intelligent Automation
- Ø Our solution is a repetitiveness and redundancy removal device
- Ø This product reduces the time approx. to 33% of the current manual labour time**

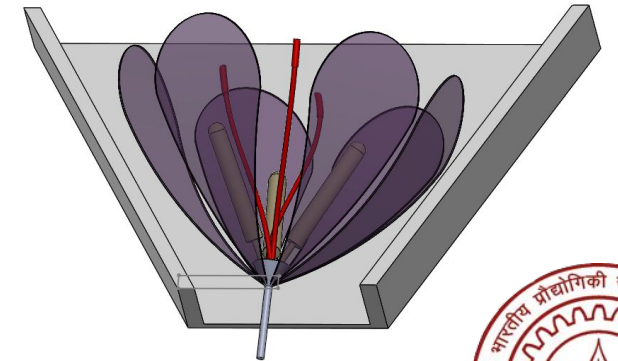
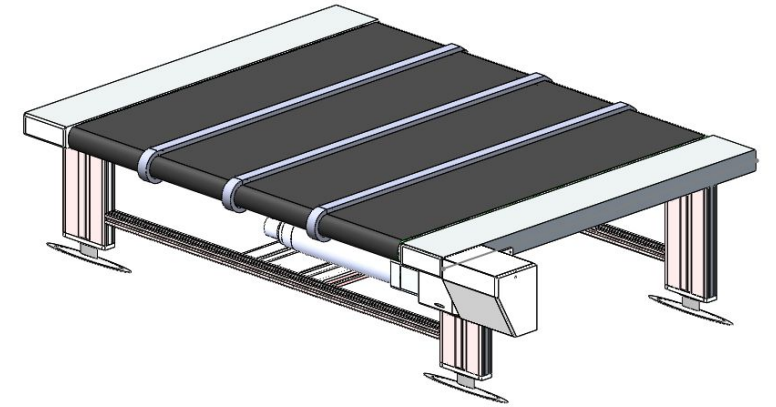


The Saffron Stigma Separator

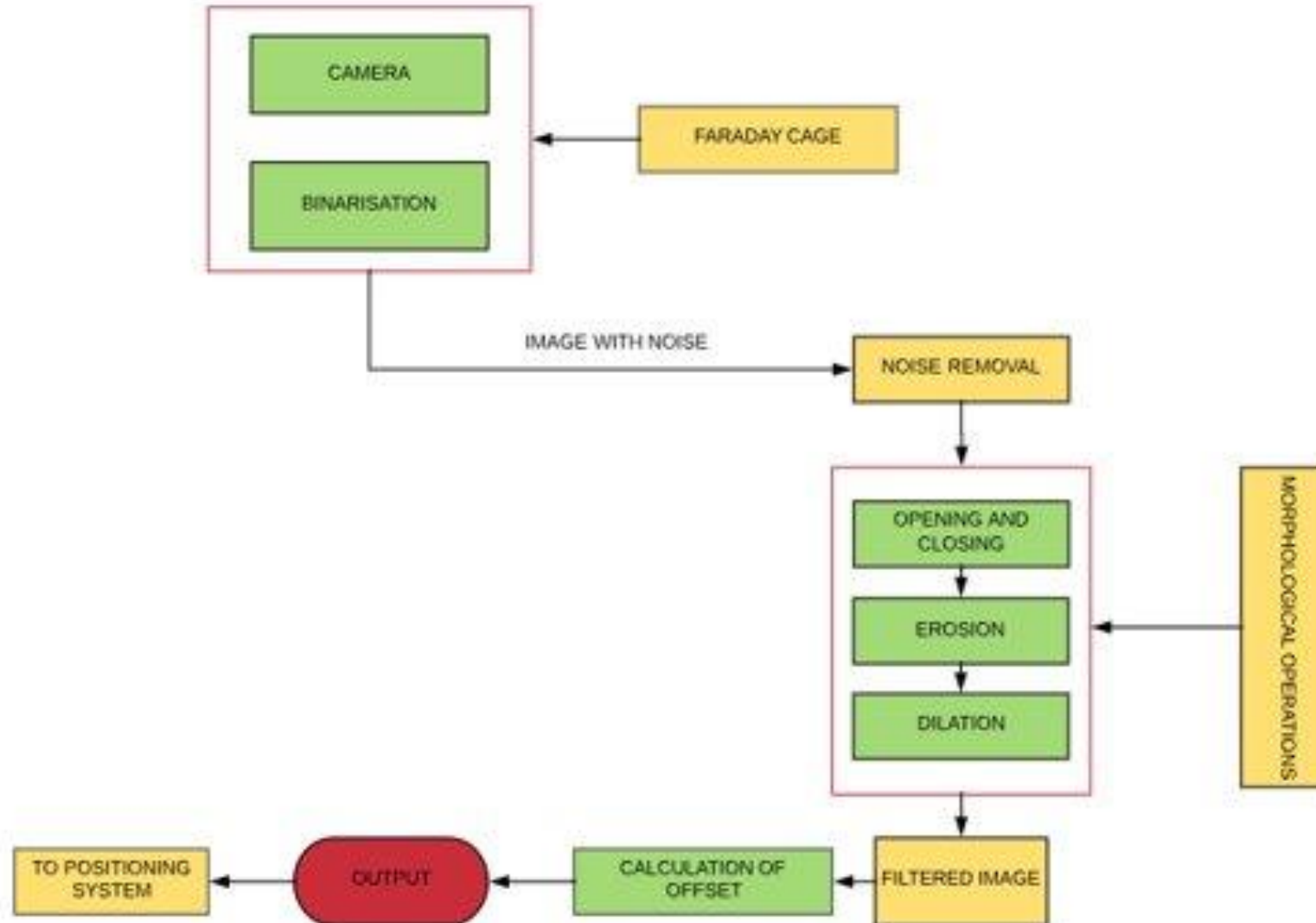


Locomotion Subsystem

- Ø Consists of a Conveyor with **4 rows** mounted on a Belt Roller assembly
- Ø Motor controls the feed rate
- Ø Transports the flower in sync to link all the subsystems
- Ø Has special **protrusions** at appropriate places to assist Image Capturing process



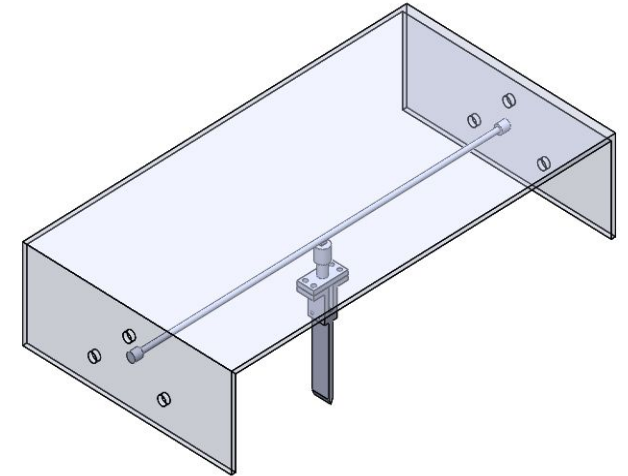
Computer Vision Subsystem



Separator Subsystem

Pneumatic based Cutter:

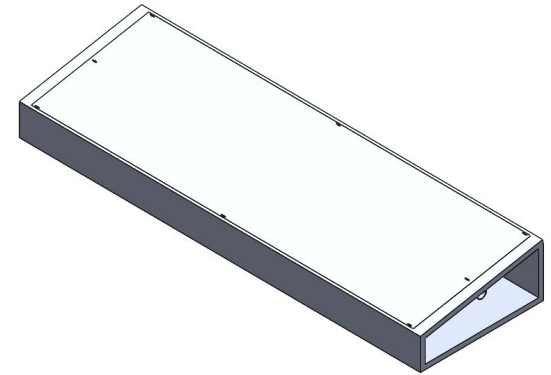
- Ø Cutter mounted on a linear positioning system
- Ø Positioning done from data acquired from CV subsystem
- Ø Pneumatic provides optimal shear force for cutting



Separator Subsystem

Vibration based Separator:

- Ø Principle: different frictional force on Petals and Stigma
- Ø Galvanized Steel sheet was chosen on the basis of maximum difference in coefficients of friction
- Ø Optimal angle, frequency and amplitude were decided based on impurities and success rate
- Ø Vibrational motors and suitable dampeners control the oscillation frequency and amplitude in practice



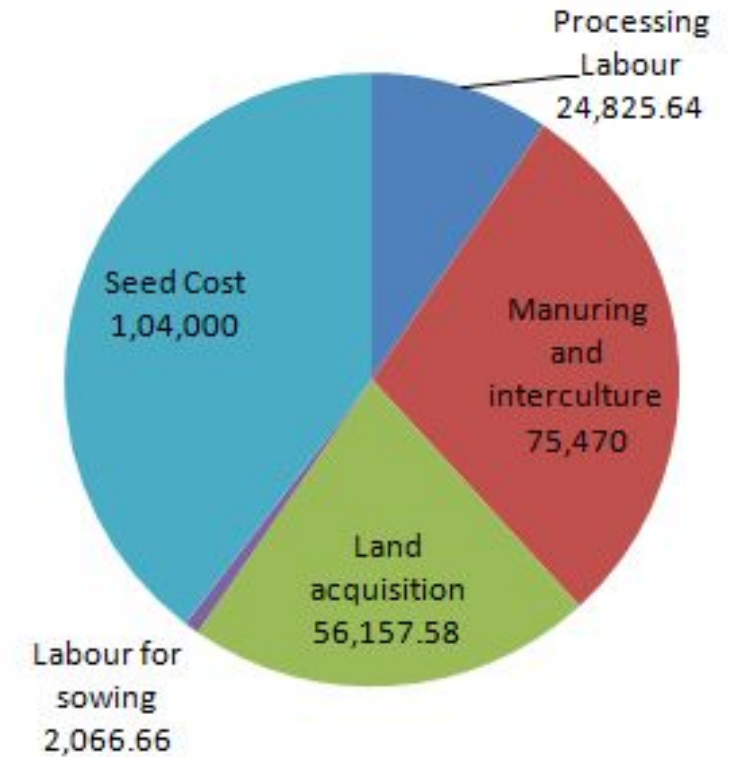
Cost Analysis of our Product

Unit	Component	Price per piece	No. of pieces	Total price
Locomotion Unit	Conveyor Belt	30-40K	1	30-40K
	Motors	5-7K	1	5-7K
	Controller(PLC)	1-2K	1	1-2K
Computer Vision Unit	Camera	4-5k	4	16-20k
	Processor	16-20k	1	16-20k
	Software*	0-20K	1	0-20k
Separator Unit	Vibration motors	5-8k	1	5-8k
	Shaker Table	3-5k	1	3-5k
Cutting Unit	Pneumatics cylinder	1-2k	4	4-8k
Separator Unit	Motors+Gears	3-4k	4	12-16k
	Controller(PLC)	1-2k	4	4-8K
	Compressed air cylinder	10-12k	1	10-12k
	Cutter blade	1-2k	4	4-8k
Assembly	Faraday Cage+Main Body	8-10k	1	8-10k
	Blade	1-2k	4	4-8k
Maintenance	Wear and Tear	5-10k		5-10k
Miscellaneous				20k
Total			Minimum	1.67 Lacs
			Maximum	2.52 Lacs



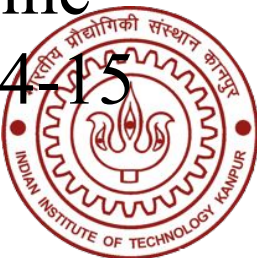
Business Case Analysis

- Processing Labour Cost of Saffron 9% of total expenditure in cultivation process
- Losses due to delayed stigma separation are as high as 8 – 12 lakhs
- A solution that process the flowers quickly as well as remove the drudgery involved will prove to be a economically viable solution

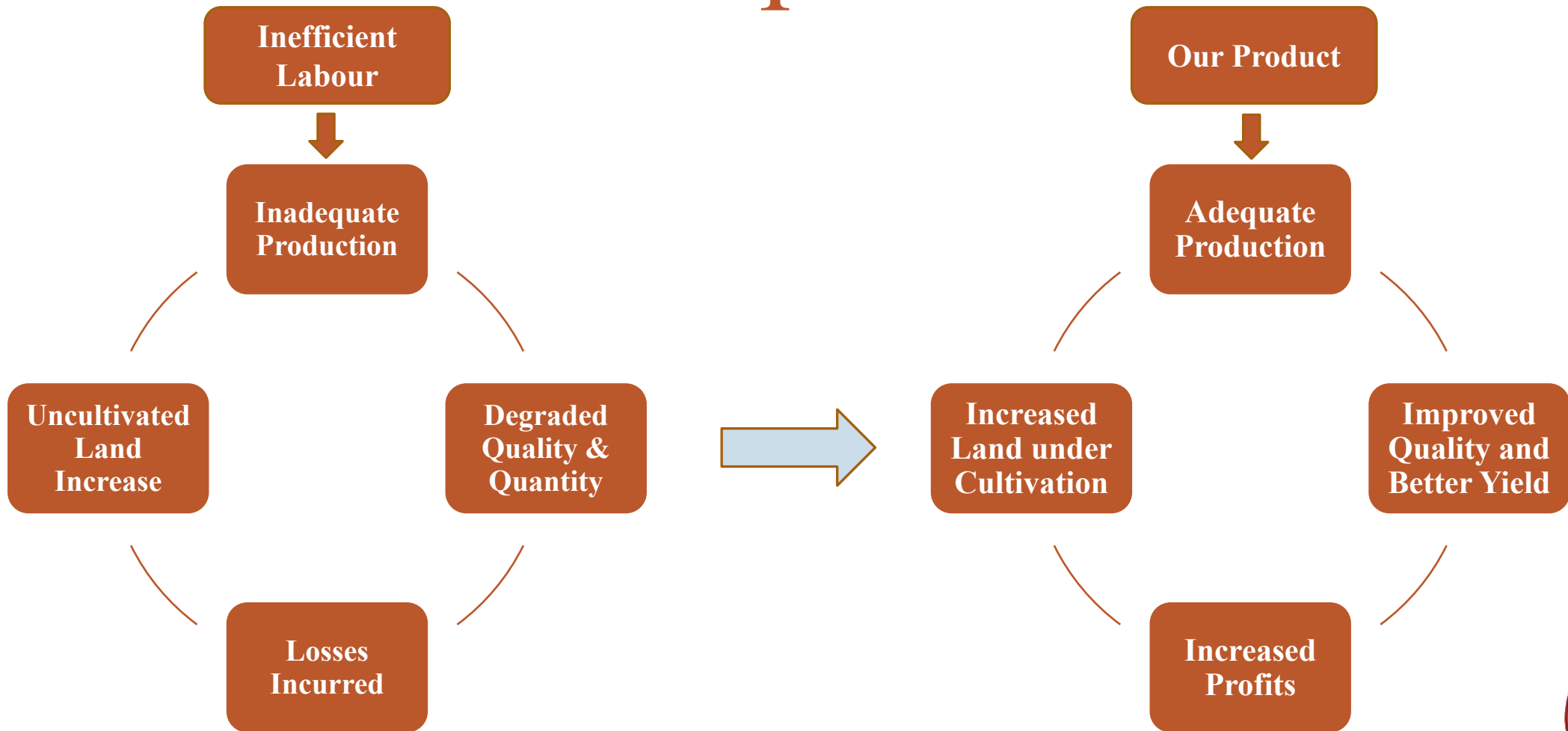


Economically Viability

- Average trained labour requires 5-6 seconds for stigma separation.
- The machine needs to process 80 flowers in a minute to cater the demand of 1 hectare field (assuming separation is done after every 3 days)
- The machine can easily process 4 flowers parallelly in 3-4 secs.
- Hence, our machine is economically viable, is reducing the labour, and also increases the productivity.
- Running cost is 0.5 lakh over a season.
- Maximum Establishment Cost is 2.5 lakhs over a season.
- Assuming that the farmer will pay the same expenditure for the machine (Rs 25k), the setup cost and running cost can be easily recovered in 14-15 days of work.



Impact!



References

- Ø 1. Luis Gracia, CarlosPerez-Vidal, Carlos Gracia-Lopez "Automated cutting system to obtain the stigmas of the saffron flower"
- Ø 2. Dr Musroor Ahmad Tantry, Professor Bashir Ahmad Dar, Dr. Sutinder singh "An Economic analysis of Production andMarketing of Saffron in Jammu and Kashmir"
- Ø 3. <http://diragriju.nic.in/saffron/saffron-manual.pdf> "Good practices for saffron production in Kashmir"- Sher-e-Kashmir University of Agricultural Sciences and Technology of Kashmir
- Ø 4. B. Emadi, and M. H. Saiedirad "Moisture-Dependent Physical Properties of Saffron Flower"
- Ø 5. Taziz Babaie1, Shamsollah Abdollahpoor, Asghar Mahmoudi & Seyyed Hossein Fattahi "Saffron Stigma Separation by Oscillating Seperator and Wind Tunnel"
- Ø 6. Amjad Masood Husaini, Badrul Hassan, Muzaffar Y. Ghani, Jaime A. Teixeira da Silva, Nayar A. Kirmani- "Saffron (Crocus sativus Kashmirianus) Cultivation in Kashmir: Practices and Problems"
- Ø 7. <https://rkvy.nic.in/static/download/pdf/Saffron Mission.pdf> -"Economic Revival of J&K saffron sector"
- Ø 8. Shabeer A. Ganaie1, Sudhakar Dwivedi and Arshad Bhat and Mehraj ud din Khanday -"A Study on Economics of Saffron (Crocus sativus L.) Cultivation in Jammu and Kashmir, India"
- Ø 9. M. Kafi and T. Showket- "A Comparative Study of Saffron Agronomy and Production Systems of Khorasan (Iran) and Kashmir (India)"



THANK YOU!!

- Any Questions ?

